

Home Assignment - 2

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Task: 1. Classes and object.

```
Class Bank {
```

```
String bankName = "SBI Bank";
```

```
void display Bank Info () {
```

```
System.out.println("Bank name: " + bank  
Name);
```

```
}
```

```
}
```

```
Class Customer {
```

```
String name;
```

```
Customer (String Name) {
```

```
this.name = name;
```

```
void display Customer () {
```

```
System.out.println(name);
```

```
}
```

```
Class Account {
```

```
int accountNo;
```

```
double balance;
```

```
Account (int accountNo, double balance) {
```

```
this.accountNo = accountNo;
```

```
this.balance = balance;
```

```
}
```

```
void display Account () {
```

```
System.out.println(accountNo, balance);
```

```
Public class main {
```

```

Public Static void main (String args[]) {
    Bank b = new Bank();
    Customer c = new Customer("Rushi");
    Account a = new Account(101, 5000);

    b.display BankInfo();
    c.display Customer();
    a.display Account();
}
}

```

Task: 2 Constructor

1. Default

```

Class Account {
    Account () {
        System.out.println("Acc created");
    }
}

```

Parameterized

```

Class Account {
    Account (int accountNo, double balance) {
        this.accountNo = accountNo;
        this.balance = balance;
    }
}

```

Copy

```

Class Account {

```

Account (int account No, double balance) {
this.account No = account No;
this.balance = balance;
}

} Account (Account a) {

this.account No = a.account No;

this.balance = a.balance;

}

}

Chaining Using this ()

Class Account {

Account () {

this (101, 333333);

Account (int account No, double balance) {

this.account No = account No;

this.balance = balance;

}

}

Task 3: Variables & Methods.

a) • Instance

Class Account {

int accountNo;

}

• local

void deposit (double amt) {

double newBal = amt;

}

• Variable

Class Bank {

Static String bankName = "ABC Bank";
}

B) Static Method

```
Class Bank {  
    Static void display Bank Info () {  
        System.out.println("ABC");  
    }  
    Bank.display Bank Info ();  
}
```

Instance Method:

```
Class Account {  
    double balance = 5000;  
    void deposit (double amt) {  
        balance + = amt;  
    }  
    void withdraw (double amt) {  
        balance - = amt;  
    }  
    void display Amount () {  
        System.out.println(balance);  
    }  
}
```

Call

```
Account a = new Account();
```

a. deposit (1000);
a. withdraw (500);
a. display Account ();

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Task - 4 Data Encapsulation.

```
class Account {
```

```
    public void setAccount (int accNo) {
```

```
        this.accNo = accNo;
```

```
    }  
    public int getAccNo () {  
        return accNo;
```

```
    }  
    public void setBalance (double bal) {  
        this.bal = bal;
```

```
    }  
    public void setBal (double bal) {  
        this.bal = bal;
```

```
    }  
    public double getBal () {  
        return bal;
```

```
    }
```

```
}
```

Task 5: Access Specifier.

	Same Class	Same Package	Subclass	Other Package
Public	✓	✓	✓	✓
Private	✓	X	X	X
Protected	✓	✓	✓	X
Default	✓	✓	X	X

Public :-

Public int accountNo;

Private :-

Private double balance;

Protected :-

protected String name;

Default :-

int age;

Task 6: Packages

Package banking;

Public class Account {

Public Account (int accNo, double bal) {

this.accNo = accNo;

this.bal = bal;

}

Public void display Account() {

```
System.out.println (accno);  
System.out.println (bal);
```

Bank.java

```
Package banking;
```

```
Public Class Bank {
```

```
    public static void display
```

```
        Bank Info () {
```

```
            System.out.println ("ABC");
```

```
        }  
    }
```

```
import banking.Account;
```

```
import banking.Bank;
```

```
public class Main {
```

```
    public static void main (String[]  
        args) {
```

```
        Bank. display Bank Info ();
```

```
        Account a = new Account (101,  
                                    3000);
```

```
        a. display Account ();
```